题目资料审核

连续浏览每道题的题目、标答和评分标准；只有编程题显示测试样例。

test-aa-ocr

新建任务 —— 上传题目 —— 3 审核题目 —— 4 上传作答 —— 5 校对作答 —— 6 执行批改 —— 7 复核批改 —— 8 结果分析

SmarTAI 智能搜索：题号、题型、题目内容，或“编程题 / 低置信 / 冲突”

第 1.1.5 题

证明题 满分 10 分 1 项需核对

筛选结果中的第 1 / 7 题

上一题 下一题

默认 10 分待确认

本题满分 请确认

系统暂按默认 10 分 · 评分标准中的步骤按百分比分配到该满分。

10 分 确认此满分 修改

题目 修改

Prove for all n > 1 that 7lf n7l is not a group under multiplication of residue classes.

标答 / 解题步骤 修改

1. For a set to be a group under multiplication, every element must have a multiplicative inverse. In the ring of residue classes modulo n (Z/nZ), an element [a] has a multiplicative inverse if and only if gcd(a, n) = 1.
2. For any n > 1, the element [0] is always present in Z/nZ. Since [0] * [a] = [0] for all [a], [0] cannot have a multiplicative inverse. Furthermore, if n is composite, there exist non-zero elements [a] such that gcd(a, n) > 1, which also lack inverses. Thus, the set fails the inverse axiom.

评分标准（与标答步骤对应） 修改

1. Identify the existence of non-invertible elements (e.g., zero divisors or elements not coprime to n): 60%. 2. Logical conclusion that the set fails the group axiom of inverses: 40%.

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第 1.1.7 题

证明题 满分 10 分 1 项需核对

筛选结果中的第 2 / 7 题

上一题 下一题

默认 10 分待确认

本题满分 请确认

系统暂按默认 10 分 · 评分标准中的步骤按百分比分配到该满分。

10 分 确认此满分 修改

题目 修改

Let G = {x E IR I 0 y x < 1 } and for x, y E G let x * y be the fractional part of x + y (i.e., x * y = x + y - [x + y] where [a] is the greatest integer less than or equal to a). Prove that * is a well defined binary operation on G and that G is an abelian group under * (called the real numbers mod 1).

标答 / 解题步骤 修改

1. Well-definedness: For any x, y in [0, 1), 0 <= x+y < 2. The fractional part {x+y} = x+y - [x+y] is either x+y (if x+y < 1) or x+y-1 (if 1 <= x+y < 2). In both cases, the result is in [0, 1), so the operation is well-defined.
2. Group Axioms: Associativity follows from the properties of the floor function. The identity is 0, since {x+0} = x. The inverse of x is 0 if x=0, and 1-x if x > 0, as {x + (1-x)} = {1} = 0.
3. Commutativity: x * y = {x+y} = {y+x} = y * x, so the group is abelian.

评分标准（与标答步骤对应） 修改

1. Proof of well-definedness: 20%. 2. Proof of group axioms (associativity, identity, inverse): 50%. 3. Proof of commutativity: 30%.

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第 1.1.20 题

证明题 满分 10 分 1 项需核对

筛选结果中的第 3 / 7 题

上一题 下一题

默认 10 分待确认

本题满分 请确认

系统暂按默认 10 分 · 评分标准中的步骤按百分比分配到该满分。

10 分 确认此满分 修改

题目 修改

For x an element in G show that x and x-t have the same order.

标答 / 解题步骤 修改

1. Let |x| = n, meaning n is the smallest positive integer such that x^n = e.
2. Consider (x^-1)^n. By the laws of exponents, (x^-1)^n = (x^n)^-1 = e^-1 = e.
3. If there existed a smaller positive integer m < n such that (x^-1)^m = e, then (x^m)^-1 = e, which implies x^m = e. This contradicts the assumption that n is the smallest positive integer such that x^n = e. Thus, |x^-1| = |x|.

评分标准（与标答步骤对应） 修改

1. Definition of order for x and x^-1: 30%. 2. Logical proof showing that x^n = 1 iff (x^-1)^n = 1: 70%.

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第 1.1.29 题

证明题 满分 10 分 1 项需核对

筛选结果中的第 4 / 7 题

上一题 下一题

默认 10 分待确认

本题满分 请确认

系统暂按默认 10 分 · 评分标准中的步骤按百分比分配到该满分。

10 分 确认此满分 修改

题目 修改

Prove that A x B is an abelian group if and only if both A and B are abelian.

标答 / 解题步骤 修改

1. If A and B are abelian, then for any (a1, b1), (a2, b2) in A x B: (a1, b1)(a2, b2) = (a1a2, b1b2) = (a2a1, b2b1) = (a2, b2)(a1, b1). Thus A x B is abelian.
2. If A x B is abelian, then (a1, e_B)(e_A, b1) = (e_A, b1)(a1, e_B) implies (a1, b1) = (a1, b1) for all a1 in A, b1 in B. Specifically, (a1, e_B)(a2, e_B) = (a1a2, e_B) = (a2a1, e_B) implies a1a2 = a2a1, so A is abelian. Similarly, B is abelian.

评分标准（与标答步骤对应） 修改

1. Proof that A and B abelian implies A x B abelian: 50%. 2. Proof that A x B abelian implies A and B abelian: 50%.

浏览态会渲染 LaTeX 或代码；点击修改可编辑源码。

第 1.1.31 题

证明题 满分 10 分 1 项需核对

筛选结果中的第 5 / 7 题

上一题 下一题

默认 10 分待确认

本题满分 请确认

系统暂按默认 10 分 · 评分标准中的步骤按百分比分配到该满分。

10 分 确认此满分 修改

题目 修改

Prove that any finite group G of even order contains an element of order 2. [Let t(G) be the set {g E G 1 g =ft g-1 }. Show that t(G) has an even number of elements and every nonidentity element of G - t(G) has order 2.]

标答 / 解题步骤 修改

1. Let t(G) = {g in G | g = g^-1}. The identity e is in t(G). If g is in t(G), then g^-1 is in t(G). Elements in t(G) other than e come in pairs {g, g^-1}. Thus, |t(G)| = 1 + 2k (odd).
2. Since |G| = |t(G)| + |G - t(G)| and |G| is even, |G - t(G)| must be odd. Since |G - t(G)| contains non-identity elements, and for any g in G - t(G), g != g^-1, these elements must pair up as {g, g^-1}. An odd number of elements cannot be fully paired, implying there must be at least one element g such that g = g^-1 (contradiction) or the logic implies an element of order 2 exists.

评分标准（与标答步骤对应） 修改

1. Analysis of the set t(G) and its cardinality: 40%. 2. Argument regarding the remaining elements in G - t(G): 60%.

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第 1.2.3 题

证明题 满分 10 分 1 项需核对

筛选结果中的第 6 / 7 题

上一题 下一题

默认 10 分待确认

本题满分 请确认

系统暂按默认 10 分 · 评分标准中的步骤按百分比分配到该满分。

10 分 确认此满分 修改

题目 修改

Use the generators and relations above to show that every element of D2n which is not a power of r has order 2. Deduce that D2n is generated by the two elements s and sr, both of which have order 2.

标答 / 解题步骤 修改

1. Elements of D2n are of the form r^k or sr^k. If an element is not a power of r, it is of the form sr^k. (sr^k)^2 = sr^k sr^k = s(r^k s)r^k = s(s r^-k)r^k = s^2 r^0 = e. Thus, all such elements have order 2.
2. Since s and sr are both not powers of r (for n > 2), they have order 2. Any element sr^k can be written as (sr)r^{k-1}, and since r = (sr)s, the set {s, sr} generates the group.

评分标准（与标答步骤对应） 修改

1. Proof that non-power-of-r elements have order 2: 60%. 2. Deduction of the generating set {s, sr}: 40%.

浏览态会渲染 LaTeX 或代码；点击修改可编辑源码。

第 1.2.16 题

证明题 满分 10 分 1 项需核对

筛选结果中的第 7 / 7 题

上一题 下一题

默认 10 分待确认

本题满分 请确认

系统暂按默认 10 分 · 评分标准中的步骤按百分比分配到该满分。

10 分 确认此满分 修改

题目 修改

Show that the group (XJ, YI I Xf = Yf = (xiYd = 1) is the dihedral group D4 (where XJ may be replaced by the letter r and y1 by s). [Show that the last relation is the same as: XJYI = YIXJ -1]

标答 / 解题步骤 修改

1. The relation (xy)^2 = 1 implies xyxy = 1. Multiplying by y^-1 on the right and x^-1 on the left gives xy = x^-1 y^-1 = (yx)^-1. Given x^2 = y^2 = 1, x = x^-1 and y = y^-1, so xy = y^-1 x^-1 = yx^-1.
2. This is the standard presentation of D4 with r=x, s=y, where r^4=1, s^2=1, and srs^-1 = r^-1. The given relations define the symmetries of a square, confirming the isomorphism to D4.

评分标准（与标答步骤对应） 修改

1. Verification of the relation equivalence: 40%. 2. Mapping to D4 presentation and isomorphism argument: 60%.

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完成全部题目资料审核

共 7 道题；确认后进入学生作答上传。

返回题目资料总览 确认全部题目资料